The Image Lives in Two Memories: Visual Information in Hybrid Linear-Attention Vision-Language Models

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Abstract

Hybrid vision-language models that interleave linear-attention layers with softmax attention read an image into two memories at once: the key-value (KV) rows of the visual tokens in the attention layers, and a fixed-size recurrent state in the linear-attention layers that has absorbed every visual token. Visual-token pruning and multimodal KV-cache compression assume that the image lives in the first of these. We ask where it actually lives. On Qwen3.5-0.8B (a Gated DeltaNet hybrid with early fusion) we alter one memory exactly once, right after the last image token, and let the question and the answer read the altered memory. A dataset-level ratio of task accuracy, anchored at a no-image run and an unaltered run, measures how much of the image's contribution each memory retains, with paired bootstrap intervals, on DocVQA, ChartQA, TextVQA and POPE; the same ratio of teacher-forced answer losses is reported alongside. [TBD: one sentence per finding: (i) KV zero vs state zero retained, per task family; (ii) bits needed by each memory after reading, with and without error feedback; (iii) fewer visual tokens versus fewer bits at equal memory; (iv) what the clean-removal controls (KV drop, state 0_C) leave.]

1 Introduction

A transformer vision-language model (VLM) keeps everything it has read about an image in one place: the keys and values of the visual tokens, one row per token and layer. Visual-token pruning [1–5] and multimodal KV-cache compression [6, 7] are built on that picture: removing or compressing a visual token's row removes or compresses what the model knows about that part of the image.

Hybrid models break the picture. Qwen3.5 [8] repeats three Gated DeltaNet (GDN) layers [9] and one gated softmax-attention layer, and it is natively multimodal: visual features enter at the input embedding and pass through the same stack as text (early fusion; Section 3.1). When the last image token has been processed, the image has been written into two memories. The attention layers hold the KV rows of the visual tokens, as in a transformer. The GDN layers hold a fixed-size recurrent state that has been updated by every visual token and is not indexed by token at all. In our setting this state is larger than the visual KV rows of any image we feed (Section 3.3).

Which of the two memories does the model consult when it answers a question about the image? The answer decides what token pruning and KV compression can safely do to a hybrid VLM, and

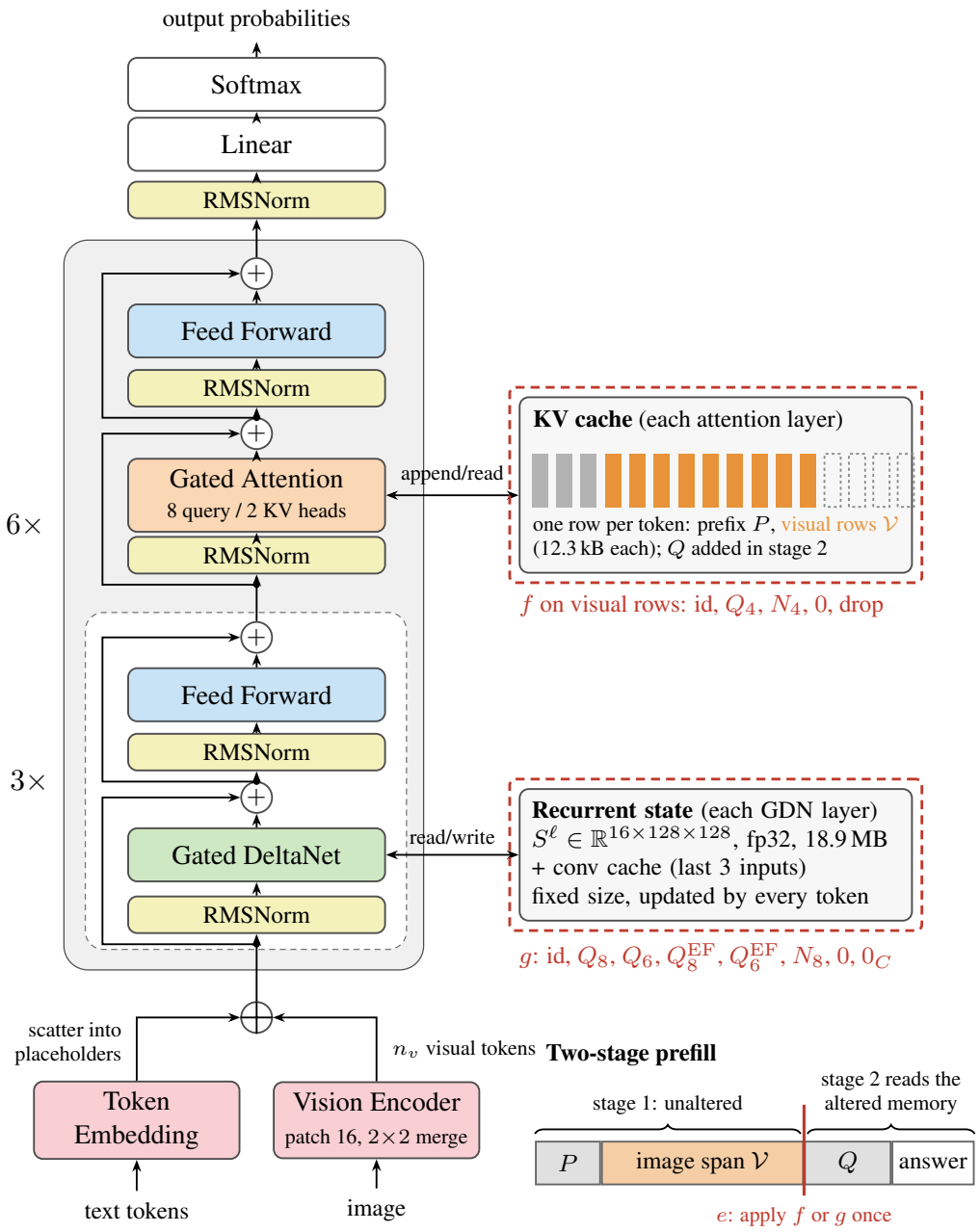


Figure 1: Qwen3.5 as a hybrid VLM and the two memories it keeps. Left: the decoder stack (pre-norm layers, six groups of three Gated DeltaNet layers and one gated attention layer for 0.8B; eight for 4B and 9B); visual tokens from the vision encoder replace placeholder tokens at the input embedding (early fusion). Right: the recurrent state of each GDN layer (fixed size, updated by every token) and the KV cache of each attention layer (one row per token, visual rows in orange). The prompt is read in two stages; right after the last image token (e) one memory is altered once, by f on the visual KV rows or by g on the state (red), and the question and the answer then read the altered memory.

whether the recurrent state, which a serving engine stores at full precision regardless of the KV format [10], is a place where image information can be lost. It is not obvious a priori. The state is written by every visual token but has a finite capacity and forgets through a learned decay; the KV rows keep every token exactly but sit in only one layer in four.

We answer the question with training-free interventions (Figure 1). The prompt is read in two stages: the prefix and the image first, then the question (Section 4.1). Between the stages we alter exactly one memory once: we zero it, quantize it, or replace it by matched noise, or we remove visual tokens before they are read. The question and the answer then read the altered memory. We measure the effect with the task accuracy of the greedy answer, normalised between a run without the image and an unaltered run (Section 4.3), so that 1 means the altered memory still supplies everything the image contributed and 0 means nothing is left; the same ratio of the teacher-forced loss of the reference answer is reported alongside.

Two text-only studies of the same model family frame the question. Split-prefill experiments on Qwen3.5 and Falcon-H1 keep only the KV cache or only the recurrent state of a prefilled context and find that exact retrieval survives only through attention, while the output language and a persona survive through the recurrent state [11]. Quantizing both memories at every write found the recurrent state to be the fragile one under long generation [12]. Neither involves images. An image differs from a text context in two ways that matter here: it is long (up to 1,280 tokens in our setting) and dense, and the questions asked about it range from reading a single word (DocVQA, TextVQA) to judging whether an object is present at all (POPE). The retrieval-versus-gist split of [11] predicts that reading tasks depend on the KV rows and presence judgements less so; our tasks test that prediction for visual content. For hybrid Mamba–Transformer video models, token reduction has been found to behave like compression rather than dropping because the state keeps information from discarded tokens [13]; we measure that state contribution directly, for still images and Gated DeltaNet.

Contributions. [TBD: Final bullets after phases 1, 2, 3 and 5. Planned:]

- A one-shot intervention protocol for hybrid VLMs that separates the two memories at the end of the image span, with a dataset-level retained ratio and paired bootstrap intervals (Sections 4.1–4.3).
- [TBD: Phase 1: which memory carries the image, per task (zero and noise interventions).]
- [TBD: Phase 2: bits each memory needs after reading; error feedback during the image prefill; one-shot versus every-write quantization.]
- [TBD: Phase 3: visual-token deletion versus KV quantization at equal memory.]
- [TBD: Phase 5: clean removal (KV drop, state 0_C): attention dilution and the convolution channel.]

2 Related Work

Visual-token reduction. FastV drops visual tokens after an early layer on the evidence of low attention to them [1]; SparseVLM ranks visual tokens with text-guided attention [2]; LLaVA-PruMerge and VisionZip select and merge tokens at the vision encoder output [3, 4]; PyramidDrop removes tokens progressively with depth [5]; token merging goes back to ToMe [14]. All of these are designed and evaluated on transformer language backbones, where the visual-token rows are the only memory of the image. For Mamba-based multimodal models, DTP scores visual tokens from the selective-scan parameters instead of attention [15]; for hybrid video models, token reduction is reported to act as compression because the state keeps what dropped tokens contributed [13].

Multimodal KV-cache compression. LOOK-M and VL-Cache compress the KV cache of multimodal inputs with modality-aware eviction and budget allocation [6, 7]. KV quantization for text, e.g. KIVI [16] and KVQuant [17], applies unchanged to the visual rows we quantize.

Where visual information flows. Attention-knockout and patching studies of transformer VLMs trace how image information reaches the answer. In InternVL2-76B and LLaVA-1.5-7B, the query tokens come to store global image information, cross-modal flow is concentrated in about a quarter of the layers in the middle of the network, and fine-grained details are read from the image tokens directly [18]. In LLaVA models, general visual features move into the question tokens in lower layers and question-relevant object information in middle layers before reaching the last position [19]; removing object-specific visual tokens lowers object identification by more than 70% [20]. Causal patching separates a direct pathway, read from the image tokens, from a text-mediated pathway through the query tokens, chosen per task, with the second acting as a fallback when the first is ablated [21]. In a hybrid model the text-mediated route has a second carrier: the recurrent state delivers image information to the question tokens without any attention to the image rows. Zeroing the visual KV rows removes the direct pathway in the attention layers and leaves this route intact; zeroing the state does the converse.

The two memories of hybrid models. Afendulev et al. separate the KV cache and the recurrent state of text contexts by keeping one and dropping the other (the convolution cache is dropped with the state) and by swapping states between contexts [11]. We differ in the input (an image span inside the prompt, followed by the question), in the interventions (graded: quantization, error feedback, matched noise and token deletion besides removal) and in the measure (a loss ratio anchored at no-image and full runs). Hybrid video models have used the state as evidence for which visual tokens to keep: as a carrier of discarded tokens [13] and, through its change over a chunk, as a query-free eviction signal for the KV cache [22].

Linear and hybrid VLMs; recurrent-state compression. Mamba-based VLMs replace the language backbone by a state-space model [23, 24]; there the state is the only memory of the image. InfiniteVL interleaves a small fraction of full-attention layers with Gated DeltaNet in a VLM built by distillation [25]; Qwen3.5 is natively pretrained as a hybrid on text and images [8]. Post-training quantization of the recurrent state of GDN and related layers has been studied for text [26–28], and jointly with the KV cache in [12]. None of these isolates the image.

3 Background

3.1 Qwen3.5 as a hybrid VLM

Qwen3.5 has L decoder layers in repeating groups of three GDN layers and one gated softmax-attention layer, so $L_a = L/4$ attention and $L_g = 3L/4$ GDN layers. For the 0.8B model we study $L = 24$, $L_a = 6$, $L_g = 18$; the 4B and 9B models have $L = 32$ (Table 1).

Early fusion. A vision encoder with patch size $p = 16$ and $m \times m$ spatial merging ($m = 2$) turns an image resized to $H \times W$ pixels (both multiples of $pm = 32$) into

$$n_v = \frac{H}{p} \cdot \frac{W}{p} \cdot \frac{1}{m^2} = \frac{HW}{(pm)^2} \tag{1}$$

visual tokens, which replace n_v placeholder tokens in the input embedding. In the transformers implementation we use, this scatter at the input embedding is the only place where visual features enter the language model; no visual features are injected at intermediate layers (Appendix A, premise 1). Visual tokens get three-dimensional rotary positions (temporal, height, width; M-RoPE [29]) on the first quarter of each attention head's dimensions. We cap the image at $HW \leq 1{,}310{,}720$ pixels, so $n_v \leq 1{,}280$.

3.2 Gated DeltaNet

Each GDN head keeps a matrix $S_t \in \mathbb{R}^{d_k \times d_v}$. With L2-normalised key $k_t \in \mathbb{R}^{d_k}$, query $q_t \in \mathbb{R}^{d_k}$ (further scaled by $d_k^{-1/2}$), value $v_t \in \mathbb{R}^{d_v}$, decay $\alpha_t \in (0, 1)$ and write strength $\beta_t \in (0, 1)$,

Table 1: Cache geometry, from the model configurations. Sizes by (4); KV in bf16, state in fp32.

Model	L	L_a	L_g	H_{kv}	d	H	$d_k{=}d_v$	$\lvert S\rvert$ (MB)
Qwen3.5-0.8B	24	6	18	2	256	16	128	18.9
Qwen3.5-4B	32	8	24	4	256	32	128	50.3
Qwen3.5-9B	32	8	24	4	256	32	128	50.3

GDN key heads: 16 in all three; the state is per value head. Query heads: 8 (0.8B), 16 (4B, 9B). Vision encoder: patch 16, merge 2; depth 12 (0.8B), 24 (4B), 27 (9B). Only 0.8B is run in this version.

computed per head from the input at t as $\alpha_t = \exp\big(-e^{A}\,\mathrm{softplus}(a_t + b)\big)$ and $\beta_t = \mathrm{sigmoid}(\beta'_t)$ with a_t, β'_t linear in the input and A, b learned, the state is first decayed and then corrected along k_t:

$$\begin{aligned} S_t &= \alpha_t S_{t-1} + \beta_t\, k_t \big(v_t - \alpha_t S_{t-1}^{\top} k_t\big)^{\top} \\ &= \alpha_t \big(I - \beta_t k_t k_t^{\top}\big) S_{t-1} + \beta_t\, k_t v_t^{\top}, \qquad (2) \\ o_t &= S_t^{\top} q_t. \qquad (3) \end{aligned}$$

This is the order of operations of the reference recurrence in the implementation (decay, read $(\alpha_t S_{t-1})^{\top} k_t$, write). Prefill computes the same recurrence in chunks, but the state is written to the cache once per forward call, at its end (premise 5). Before the recurrence, q, k, v pass through a causal depthwise convolution of kernel size $\omega = 4$ whose cache holds the last inputs; we return to it in Section 4.5.

3.3 Two memories and their sizes

With H_{kv} KV heads of dimension d per attention layer and H GDN value heads, a sequence of T tokens leaves

$$|\mathrm{KV}|(T) = 2 L_a H_{kv} d\, T, \qquad |S| = L_g H d_k d_v \qquad (4)$$

elements in the two memories (the factor 2 counts keys and values). The implementation stores the KV cache in the model dtype (bf16, 16 bits) and the recurrent state in fp32 (32 bits; premise 3). For 0.8B, $|\mathrm{KV}|(T) = 6{,}144\,T$ and $|S| = 18 \cdot 16 \cdot 128 \cdot 128 = 4{,}718{,}592$, i.e. 12.3 kB of KV per token and an 18.9 MB state. The visual rows of an image therefore occupy $0.0123\,n_v$ MB, at most 15.7 MB at $n_v = 1{,}280$, and the two are equal only at

$$n_v^{\star} = \frac{32\,|S|}{16 \cdot 2 L_a H_{kv} d} = 1{,}536 \qquad (5)$$

visual tokens. The 4B and 9B models have both memories $8/3$ times larger (32.8 kB per token, 50.3 MB) and the same crossover. Every image in this study leaves a state larger than its own KV rows. (The crossover of 768 tokens in [12] counts the state in bf16; the implementation used here keeps it in fp32.)

4 Method

4.1 Two-stage prefill and the intervention point

A prompt is the token sequence $x = (P, \mathcal{V}, Q)$: a prefix P (chat template up to and including the vision-start token), the image span $\mathcal{V} = \{s, \ldots, e-1\}$ of $n_v = e - s$ placeholder positions carrying the visual features, and the question Q (vision-end token, question, instruction and the assistant header). Let $\mathcal{M}_e$ be the state of all caches after the first e tokens:

$$\mathcal{M}_e = \big(\{K^{\ell}_{1:e}, V^{\ell}_{1:e}\}_{\ell \in \mathcal{A}},\ \{S^{\ell}_e,\, C^{\ell}_e\}_{\ell \in \mathcal{G}}\big), \qquad (6)$$

with $\mathcal{A}$, $\mathcal{G}$ the attention and GDN layers and C^{ℓ}_e the convolution cache. Stage 1 runs $(P, \mathcal{V})$ with unaltered computation. An intervention ϕ maps $\mathcal{M}_e$ to $\phi(\mathcal{M}_e)$. Stage 2 runs Q on $\phi(\mathcal{M}_e)$ without

further alteration; the reference answer is then scored by teacher forcing on a copy of the resulting cache, and a greedy answer of at most 16 tokens is decoded on the original. Because nothing is altered after the image, any effect of ϕ is an effect on what the image left behind. All conditions, including the unaltered one, use the same two-stage path. [TBD: two-stage versus single-pass gold NLL on the first 50 samples of each task (scripts/check_twostage.py, D-32).]

4.2 Interventions

KV side. For every attention layer, the rows of the visual positions are replaced, $K_{\mathcal{V}}^{\ell} \leftarrow f(K_{\mathcal{V}}^{\ell})$ and $V_{\mathcal{V}}^{\ell} \leftarrow f(V_{\mathcal{V}}^{\ell})$, with $f \in \{\mathrm{id}, Q_4, N_4, 0\}$, or the visual rows are removed from the cache altogether (*drop*). Rows of P are not touched, and rows of Q are written later by stage 2. After a drop, the causal-mask positions of later tokens follow the shorter cache while their rotary positions are unchanged, so the result equals keeping the rows and masking their attention columns (verified to 10^{-4} on a test model). Keys are cached after the rotary embedding, so f acts on rotated keys.

State side. For every GDN layer, $S_e^{\ell} \leftarrow g(S_e^{\ell})$ with $g \in \{\mathrm{id}, Q_8, Q_6, Q_8^{\mathrm{EF}}, Q_6^{\mathrm{EF}}, N_8, 0\}$; these leave the convolution cache C_e^{ℓ} in place. A last level, 0_C, zeroes both S_e^{ℓ} and C_e^{ℓ}, so that nothing of the image survives in the GDN layers.

Quantizer. Q_b is asymmetric uniform round-to-nearest with one scale and zero point per group of $G = 128$ consecutive elements along the last dimension (the head dimension $d = 256$ for keys and values, two groups per head and token; d_v for the state, one group per row $S[h, i, :]$). For a group $x \in \mathbb{R}^G$,

$$\begin{aligned} \sigma &= \frac{\max x - \min x}{2^b - 1}, \qquad z = \left\lfloor -\min x/\sigma \right\rceil, \\ Q_b(x) &= \sigma\Big(\mathrm{clamp}\big(\lfloor x/\sigma \rceil + z,\, 0,\, 2^b - 1\big) - z\Big), \end{aligned} \tag{7}$$

where $\lfloor \cdot \rceil$ rounds to the nearest integer and σ is floored at 10^{-8}. Quantization is simulated in fp32 and cast back to the cache dtype. Because the grid spans the group's range, every element is within half a step of its stored value, $|Q_b(x)_i - x_i| \le \sigma/2$, i.e. at most $1/(2(2^b - 1))$ of the group range: 3.3% at 4 bits, 0.79% at 6 bits and 0.20% at 8 bits.

Matched noise. N_b adds Gaussian noise with the energy of b-bit quantization error but none of its structure. For a cache tensor X (one layer's visual keys, values, or state),

$$N_b(X) = X + \rho_b(X)\, Z, \quad \rho_b(X) = \sqrt{\tfrac{1}{|X|} \textstyle\sum \big(Q_b(X) - X\big)^2}, \tag{8}$$

with Z standard normal entries of the shape of X. The generator is re-seeded (seed 0) for every tensor, so tensors of the same shape receive the same Z, across layers and samples, scaled by their own ρ_b. The RMS ρ_b is taken over the whole tensor, so noise is spread uniformly where quantization error is concentrated in high-range groups.

Error feedback during the image. Quantizing the state once after the image (Q_b above) commits one rounding error. With error feedback the image is read in chunks of at most 256 visual tokens, with boundaries $s = e_0 < e_1 < \cdots < e_C = e$, $C = \lceil n_v/256 \rceil \le 5$. Let $\hat{S}_{e_u}$ be the state after running chunk u from the stored state $\tilde{S}_{e_{u-1}}$, with $\tilde{S}_{e_0} = S_s$ the unaltered state after the prefix. Then

$$\tilde{S}_{e_u} = Q_b\big(\hat{S}_{e_u} + r_{u-1}\big), \quad r_u = \hat{S}_{e_u} + r_{u-1} - \tilde{S}_{e_u}, \tag{9}$$

with $r_0 = 0$; the final residual r_C is discarded and $\tilde{S}_e$ is what the question reads. Each chunk is a separate forward call, because the state is written once per call (premise 5). For $n_v \le 256$, $C = 1$ and $Q_b^{\mathrm{EF}} = Q_b$. Error feedback here therefore acts across at most five stores, unlike decoding in [12], where it acts across every generated token.

Zeroing and removal. $f = 0$ sets visual keys and values to zero and $f =$ drop removes them; $g = 0$ sets every GDN state to zero and $g = 0_C$ also its convolution cache. Section 4.5 states what each removes and what it leaves; the pairs (0, drop) and $(0, 0_C)$ separate the two side effects from the removal itself.

Visual-token deletion. A keep set $I \subset \{0, \ldots, n_v - 1\}$ of size $n'_v = \max(1, \mathrm{round}(\kappa n_v))$, $\kappa \in \{0.5, 0.25\}$, is chosen uniformly, $I = \{\lfloor j\, n_v / n'_v \rfloor : j = 0, \ldots, n'_v - 1\}$, or at random without replacement (seed 0). The other visual tokens are removed from the input before the language model, so neither memory ever sees them; the kept tokens keep their original M-RoPE positions, and the vision encoder still sees the full image. Deletion shrinks the visual KV rows to n'_v and leaves $|S|$ unchanged.

Every-write quantization. For comparison with [12] we also quantize every KV append and every state write of the whole run with the released hooks of that work (scope *whole*); the image and the text are quantized alike. The state is then stored at the end of stage 1, at the end of stage 2, after the teacher-forced pass over the reference answer, and after every decoded token. With error feedback the residual is reset at the first store of each layer (end of stage 1) and carried through the later stores. The residual buffer is shared between the teacher-forced copy of the cache and the decoding cache, so the first decoding step continues from the residual of the teacher-forced pass; this affects the greedy answers of the two every-write cells with error feedback, not their losses. We report these two cells as run.

4.3 Retained ratio

We report two retained ratios with the same anchors. Two anchors use the same prompt: *full* (image, no alteration) and *blind* (no image: the image span and its start and end tokens are removed).

Accuracy retained (primary). With $\bar{s}(c)$ the mean task score (defined below) of the greedy answers to a task's n samples under condition c,

$$\mathcal{R}(c) = \frac{\bar{s}(c) - \bar{s}(\mathrm{blind})}{\bar{s}(\mathrm{full}) - \bar{s}(\mathrm{blind})}. \tag{10}$$

$\mathcal{R} = 1$: the altered memory supplies the full gain in task score that the image provides; $\mathcal{R} = 0$: none of it; $\mathcal{R} < 0$: worse than no image. It is defined when $\bar{s}(\mathrm{full}) > \bar{s}(\mathrm{blind})$, which we check per task before reporting.

Loss retained (secondary). Let the reference answer have tokens $a_{1:m}$. For sample i under condition c,

$$\mathrm{NLL}_i(c) = -\frac{1}{m} \sum_{j=1}^{m} \log p_c\big(a_j \mid x,\, a_{<j}\big), \tag{11}$$

where p_c reads the memory produced by condition c. With $\bar{N}(c) = \frac{1}{n} \sum_i \mathrm{NLL}_i(c)$ over a task's n samples,

$$\mathcal{R}_{\mathrm{NLL}}(c) = \frac{\bar{N}(\mathrm{blind}) - \bar{N}(c)}{\bar{N}(\mathrm{blind}) - \bar{N}(\mathrm{full})}, \tag{12}$$

read as $\mathcal{R}$ with the reduction in answer loss in place of the gain in score. Both ratios are ratios of dataset means, not means of per-sample ratios, whose denominators can be near zero (for scores in $\{0, 1\}$ they are undefined for most samples). Writing B_i, F_i, C_i for the three per-sample losses,

$$\begin{aligned} \mathcal{R}_{\mathrm{NLL}}(c) &= \frac{\sum_i (B_i - C_i)}{\sum_i (B_i - F_i)} = \sum_i \omega_i \, \frac{B_i - C_i}{B_i - F_i}, \\ \omega_i &= \frac{B_i - F_i}{\sum_{i'} (B_{i'} - F_{i'})}, \end{aligned} \tag{13}$$

where the second equality requires $B_i \neq F_i$ for every sample. $\mathcal{R}_{\mathrm{NLL}}$ thus weights each sample by how much the image helps it: samples the image barely helps contribute little, and samples it hurts ($B_i < F_i$) enter with negative weight. Every sample's loss is a per-token mean, so long answers do not dominate. It is defined when $\bar{N}(\mathrm{blind}) > \bar{N}(\mathrm{full})$, which we check per task before reporting. For both ratios the 95% interval is a paired percentile bootstrap [30]: 2,000 resamples of sample indices with replacement (seed 0), the same indices for the three means; resamples with a zero denominator are dropped.

Reference answers and task scores. The reference for the loss is the first answer (DocVQA), the most frequent of the ten annotator answers (TextVQA), or the single answer (ChartQA, POPE). Greedy answers are scored with ANLS at $\tau = 0.5$ [31] (DocVQA), relaxed accuracy with 5% relative tolerance [32, 33] (ChartQA), VQA accuracy $\frac{1}{10}\sum_{i=1}^{10}\min\big(1, \frac{1}{3}\#\{j \neq i : a^{(j)} = \hat{a}\}\big)$ [34] (TextVQA), and yes/no accuracy and F1 with *yes* as the positive class (POPE), all after one shared normalisation (lower case; punctuation except decimal points and percent signs removed; articles removed).

Why accuracy is primary. The loss is taken on the reference string as released, including its letter case, whereas the scores are computed after lower-casing. The two disagree where the model's own casing differs from the reference: in the unaltered condition, the greedy answer differs from the reference only in letter case for 37% of POPE samples (references are lower-case *yes*/*no*, the model writes *Yes*/*No*) and 20% of DocVQA samples, against under 1% for ChartQA and TextVQA. On those samples NLL_i measures the cost of an unusual casing as well as of a wrong answer, so a condition that only shifts casing moves $\mathcal{R}_{\mathrm{NLL}}$ and not $\mathcal{R}$; $\mathcal{R}_{\mathrm{NLL}}$ can then exceed 1 or change sign while the answers stay correct. We therefore draw conclusions from $\mathcal{R}$, report $\mathcal{R}_{\mathrm{NLL}}$ alongside, and mark the POPE and DocVQA columns of $\mathcal{R}_{\mathrm{NLL}}$.

4.4 Memory accounting

For the equal-memory comparison we charge a condition

$$\mathrm{MB}(c) = \frac{b_{kv}\,|\mathrm{KV}|(n'_v + n_t) + b_s\,|S|}{8 \cdot 10^6}, \tag{14}$$

with n'_v the visual tokens kept, n_t the text tokens, $b_{kv} \in \{16, 4, 0\}$ (full, 4-bit, zero) and $b_s \in \{32, 8, 6, 0\}$; noise is charged at its matched bit-width. This is deployment accounting: all KV rows are charged at b_{kv}, although the one-shot intervention leaves the text rows at 16 bits. Scales and zero points (about $32/G = 0.25$ bits per element for fp16 pairs) and the convolution cache are not counted.

4.5 Caveats of the interventions

Zeroed keys still take attention. A zero key gives the logit $q^\top 0/\sqrt{d} = 0$, not $-\infty$, so each zeroed visual position keeps a softmax weight $w_i = 1/\mathcal{Z}$, with $\mathcal{Z}$ the softmax normaliser of the query, and contributes the zero value. The attention output becomes $\sum_{i \notin \mathcal{V}} w_i v_i$ with $\sum_{i \notin \mathcal{V}} w_i = 1 - n_v/\mathcal{Z} < 1$: KV zero removes the visual content and also shrinks the attention output by the mass the zeroed rows absorb. Masking the visual positions would remove the content and renormalise, which is what the drop level does; [11] avoid zero keys for this reason. The difference between KV zero and KV drop measures the dilution.

The convolution cache leaks the last visual tokens. Unlike [11], who drop the convolution cache together with the state, we leave it in place, following [12]. C_e^ℓ holds the inputs of the last positions of the span, and the causal convolution of kernel $\omega = 4$ mixes them into the GDN inputs of the first $\omega - 1 = 3$ tokens after the span. State zero therefore leaves a short local channel from the last three visual tokens into the first question-side tokens (and, through their hidden states, into everything above them). The 0_C level removes this channel, and its difference from state zero measures it.

Table 2: Carriers of image information after each intervention. ✓ intact, ∼ degraded, – removed. Conv: the last three visual tokens through the convolution cache.

Intervention	visual KV	state	conv
full	✓	✓	✓
KV Q_4, N_4	∼	✓	✓
KV 0	– (dilution)	✓	✓
state Q_b, Q_b^{EF}, N_8	✓	∼	✓
state 0	✓	–	✓
KV drop	–	✓	✓
state 0_C	✓	–	–
KV 0 + state 0	– (dilution)	–	✓
KV drop + state 0_C	–	–	–
deletion, keep κ	κn_v rows	κn_v tokens	✓

What each intervention leaves. Table 2 lists the carriers of image information left after each intervention. The prefix rows of the KV cache precede the image and carry none of it. Zeroing both memories leaves only the convolution channel and the attention dilution of the zeroed keys, so $\mathcal{R}$ of that cell bounds what these two side effects contribute. KV drop with state 0_C leaves no carrier at all; its $\mathcal{R}$ should be close to 0, and it serves as a check of the protocol (it differs from the blind anchor only in that the prompt keeps the vision start and end tokens and the original positions).

State zero also forgets the prefix. S_e^ℓ summarises P as well as $\mathcal{V}$; zeroing it removes both. The prefix is the fixed chat-template header, the same for every sample.

4.6 Predictions

We state what prior work predicts before reporting results. (P1) If exact retrieval goes through attention, as in text [11], then reading tasks (DocVQA, TextVQA, and the values read off charts in ChartQA) should lose most of the image's contribution under KV zero and little under state zero. (P2) A coarse judgement such as object presence (POPE) may survive through the state alone, as the gist of a text context does [11]. (P3) In the one-shot cells a single rounding error is injected after the image and then carried, and damped by the decay, through the unquantized updates of the question tokens; the repeated injection that made the state fragile under every-write quantization [12] does not occur, so their loss should be well below that of the every-write cells at the same bit-widths. (P4) Error feedback can only matter for images with $n_v > 256$ ($C > 1$). [TBD: keep, revise or drop each prediction against the results; report P1–P4 outcomes explicitly.]

5 Experimental Setup

Models. Qwen3.5-0.8B in all phases run (1, 2, 3 and 5), bf16 weights, the post-trained (non-base) checkpoint. The planned 9B phase (phase 4) is not run in this version. On the MPS backend neither the fla kernels nor the fused causal convolution are available, so GDN and its convolution run in the library's PyTorch reference implementation; transformers 5.22.2 and torch 2.15.0, as recorded in every result row.

Data. Fixed random subsets (seed 0) of DocVQA validation (200) [35], ChartQA test (200) [33], TextVQA validation (200) [36] and POPE (300) [37], from the lmms-lab copies used by lmms-eval [38] at pinned revisions; the sample ids are released. Every question gets the same instruction, "Answer the question using a single word or phrase.", in the chat template without a system prompt, rendered with `enable_thinking=False` so that the prompt closes an empty thinking block before the answer (non-thinking mode); the code rejects any prompt that ends in an open thinking block, since the 4B and 9B templates open one by default.

Conditions. Phase 1 (which memory): $f \in \{Q_4, N_4, 0\}$ with the state unaltered; the state levels Q_8, Q_8^{EF}, Q_6, Q_6^{EF}, N_8 and 0 with the KV unaltered; and both zeroed. Phase 2 (bits): the state levels alone and with $f = Q_4$, and the two every-write cells (KV4 with Q_8 and Q_8^{EF}). Phase 3 (tokens versus bits): $\kappa \in \{1, 0.5, 0.25\}$ uniform, with KV at 16 or 4 bits, and $\kappa = 0.5$ random. Phase 5 (clean removal): KV drop, state 0_C, and both. (Phase 4, five conditions on 9B, is not run.) Each condition also has the two anchors per task. Batch size 1 and greedy decoding throughout.

Compute. One Apple M3 Pro laptop with 18 GB of unified memory (MPS backend); no cloud GPU. Phase 1 took 0.73 s per sample on average over 10,800 samples (Appendix C). [TBD: total hours of phases 2, 3 and 5.]

6 Results

6.1 Which memory carries the image

[TBD: Figure: retained per condition and task with 95% CIs (paper/figures/retained_Qwen3.5-0.8B.pdf; $\mathcal{R}$ and $\mathcal{R}_{\mathrm{NLL}}$ side by side). Table: phase-1 retained. Text: KV zero vs state zero; KV noise vs KV4; state noise vs state 8 (structure vs energy, cf. the noise control in [12]); both zeroed versus blind (should be near 0 unless the conv leak carries information); phase 5: KV drop vs KV zero (dilution), state 0_C vs state zero (conv channel), KV drop + state 0_C (protocol check, $\mathcal{R} \approx 0$).]

6.2 How many bits each memory needs after reading

[TBD: State 8/6 with and without EF, alone and with KV4; P3 (one-shot versus every-write at KV4 with Q_8 and Q_8^{EF}); P4 (split EF effects by C, with the n_v distribution per task from the result rows).]

6.3 Fewer visual tokens or fewer bits

The image's KV rows cost $\kappa\, b_{kv}$ bits per element and original visual token: 16, 8, 4 at $\kappa = 1, 0.5, 0.25$ with 16-bit KV, and 4, 2, 1 with 4-bit KV. The equal-memory pair is therefore $\kappa = 0.25$ at 16 bits against $\kappa = 1$ at 4 bits (4 bits each); the state (32 bits) is the same in both, and the text rows differ (16 versus 4 bits), which (14) charges. [TBD: retained and task scores of the pair; the full $\kappa \times b_{kv}$ grid as a frontier against MB; uniform versus random at $\kappa = 0.5$.]

6.4 Task scores

[TBD: ANLS / relaxed accuracy / VQA accuracy / POPE F1 per condition; does the ordering agree with $\mathcal{R}$?]

7 Discussion and Limitations

[TBD: Practical reading for pruning and KV compression of hybrid VLMs.]

Limitations. (i) KV zero keeps softmax mass on the zeroed rows and state zero leaves the convolution cache (Section 4.5); the drop and 0_C levels remove these side effects. (ii) Each intervention acts once, after the image; it does not model a cache that stays quantized while the answer is generated, except for the two every-write cells. (iii) $\mathcal{R}_{\mathrm{NLL}}$ uses the loss of one reference answer per sample, verbatim and with its letter case, so a reference that the model would phrase or case differently inflates both anchors (Section 4.3); $\mathcal{R}$ avoids this but is coarser, since each sample's score is close to binary. (iv) Removing a pathway can show what the model is able to fall back on rather than what it normally uses [21]: $\mathcal{R}$ under KV zero or state zero measures the information the other memory can supply on its own, not how much of it the intact model draws from each. (v) Subsets of 200–300 samples per task, a single model (0.8B), one image resolution cap; whether the 4B and 9B models, whose memories are $8/3$ times larger, split the image the same way is not tested. (vi)

Quantization is simulated; no memory or speed is saved in these runs. (vii) All runs use the PyTorch reference kernels on MPS; bf16 numerics may differ slightly from the fused CUDA kernels used in deployment. Repeating the core conditions on 4B and 9B, where the state and KV are larger but the crossover is the same, is left to future work.

8 Conclusion

[TBD: after results.]

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Reproducibility and Data Availability

Code, configurations, subset ids, raw per-sample results and the scripts that regenerate every table and figure are at `https://github.com/skytinstone/twomemories` [TBD: make public at submission]. Each phase is run by `python scripts/run_phase.py --phase <name>`, the tables by `python analysis/make_tables.py` and the figures by `python analysis/figures.py`. Quantization uses the released `hybridquant` package [12] unchanged. All datasets are public.

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A Implementation premises

Checked against transformers 5.22.2 (Qwen3.5 modeling and cache code) before any run; file and line references are in `docs/assumptions.md` of the repository.

	Premise	Consequence for the method
1	Visual features are scattered into the input embedding only; no intermediate-layer injection.	The caches after the span (KV, state, convolution cache) are the only carriers of the image.
2	The image is one contiguous span of placeholder tokens of length $\prod(\text{grid})/m^2$.	$\mathcal{V}$ is known exactly; (1).
3	The recurrent state is stored in fp32 (PyTorch path).	State charged at 32 bits; (5). The PyTorch path is the one used in all runs.
4	The cache is a `DynamicCache` with attention and linear-attention layers; the write methods hooked by `hybridquant` exist unchanged.	No adapter; every-write cells use the hooks as released.
5	Prefill computes GDN in 64-token chunks internally but writes the state once per forward call.	Error feedback during the image needs one forward call per 256-token chunk (9).

B Conditions

The 22 conditions of phases 1–3 and the three of phase 5, run on Qwen3.5-0.8B, with the phases that run them; the five marked P4 form the planned 9B phase, which is not run in this version. Bits are the charges of (14). Every task also has the blind anchor.

Label	KV f	state g	tokens	scope	phases	b_{kv}	b_s
KVfull_Sfull	id	id	all	image	P1–P4	16	32
KV4_Sfull	Q_4	id	all	image	P1, P2, P3	4	32
KVnoise_Sfull	N_4	id	all	image	P1	4	32
KVzero_Sfull	0	id	all	image	P1, P4	0	32
KVfull_S8	id	Q_8	all	image	P1, P2	16	8
KVfull_S8ef	id	Q_8^{EF}	all	image	P1, P2	16	8
KVfull_S6	id	Q_6	all	image	P1, P2	16	6
KVfull_S6ef	id	Q_6^{EF}	all	image	P1, P2	16	6
KVfull_Snoise	id	N_8	all	image	P1	16	8
KVfull_Szero	id	0	all	image	P1, P4	16	0
KVzero_Szero	0	0	all	image	P1	0	0
KV4_S8	Q_4	Q_8	all	image	P2	4	8
KV4_S8ef	Q_4	Q_8^{EF}	all	image	P2, P4	4	8
KV4_S6	Q_4	Q_6	all	image	P2	4	6
KV4_S6ef	Q_4	Q_6^{EF}	all	image	P2	4	6
KV4_S8_whole	Q_4	Q_8	all	whole	P2	4	8
KV4_S8ef_whole	Q_4	Q_8^{EF}	all	whole	P2	4	8
KVfull_Sfull_T50u	id	id	50% uniform	image	P3, P4	16	32
KV4_Sfull_T50u	Q_4	id	50% uniform	image	P3	4	32
KVfull_Sfull_T25u	id	id	25% uniform	image	P3	16	32
KV4_Sfull_T25u	Q_4	id	25% uniform	image	P3	4	32
KVfull_Sfull_T50r	id	id	50% random	image	P3	16	32
KVdrop_Sfull	drop	id	all	image	P5	0	32
KVfull_Szeroconv	id	0_C	all	image	P5	16	0
KVdrop_Szeroconv	drop	0_C	all	image	P5	0	0

C Compute

One Apple M3 Pro laptop (18 GB unified memory, MPS backend). Seconds are summed per-sample wall time of the evaluation (two-stage prefill, teacher-forced pass, greedy decoding) from `results/mac/compute_log.jsonl`, excluding model loading and data download. Cells shared between phases are run once and counted in the first phase that runs them. The process footprint, including MPS memory, stayed near 8 GB as read by the live monitor; it is not recorded in the result rows.

Run	Model	samples	s/sample	hours
phase 1	0.8B	10,800	0.73	2.20
phase 2	0.8B	[TBD:]		
phase 3	0.8B	[TBD:]		
phase 5	0.8B	[TBD:]		

D Loss and accuracy on POPE

[TBD: examples where $\mathcal{R}_{\mathrm{NLL}}$ and $\mathcal{R}$ disagree (state zero: $\mathcal{R}_{\mathrm{NLL}} = -0.65$, accuracy kept), with the casing of the greedy answers per condition; figure from analysis/figures.py.]

E Full results

[TBD: generated by analysis/make_tables.py from results/**/*.jsonl.]